

RAKSHITA VIJAY

Hyderabad, India | f20232010@hyderabad.bits-pilani.ac.in
GitHub | LinkedIn

SUMMARY

I'm a third-year Computer Science Engineering student at BITS Pilani, Hyderabad, with a habit of picking projects apart down to the data structure underneath — whether that's a dependency graph deciding the order database objects get migrated in, or a function-call graph deciding whether an Android app is malicious. Most recently I was a Software Engineering Intern on Microsoft's Azure Data Migration team, where I extended an AI-assisted schema-conversion pipeline to handle SQL Server as a source: a topological chunk scheduler, a hybrid regex-plus-LLM translation router, and a fault-tolerant compile-and-retry loop that got ~97% of chunks converting without manual intervention. Before that, at CultureVo, I built a multi-agent conversational AI toolkit spanning six personas and four languages.

EDUCATION

BITS Pilani, Hyderabad Campus <i>B.E. Computer Science Engineering · CGPA 8.64</i>	<i>2023 — 2027</i>
St. John's Public School <i>CBSE · 94.2%</i>	<i>Class XII · 2023</i>
Chettinad Vidyashram <i>CBSE · 95.8%</i>	<i>Class X · 2021</i>

TECHNICAL SKILLS

Python, C++, C, Java, SQL, PyTorch, Next.js, React, Supabase, Vercel, CrewAI, Gemini API, Prompt Engineering, Git

EXPERIENCE

Microsoft — Azure Data Migration Team <i>Software Engineering Intern · SQL Server → PostgreSQL Schema Conversion Engine</i>	<i>May 2026 — Jul 2026</i>
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- Extended an AI-assisted Oracle→PostgreSQL schema-conversion pipeline to support SQL Server as a source, with dialect-aware gating across a 4-stage Extract→Chunk→Convert→Verify pipeline.
- Built a dependency-aware chunk scheduler on Kahn's Algorithm to topologically order database objects, correctly handling cycles and self-referencing edges while upholding ACID transactionality.
- Designed a hybrid conversion router splitting mechanical DDL to a regex fast-path and procedural/semantic logic to LLM translation, sharing a canonical identifier map across both paths.
- Built a fault-tolerant compile-and-retry loop with auto-retry from compile-error feedback — ~97% of chunks converged without manual intervention.
- Stress-tested against Northwind, AdventureWorks, WideWorldImporters, and a custom 7-schema database: ~98% of in-scope objects across 45+ chunks (600+ objects), cutting runtime from ~4 hours to 35–45 minutes.

CultureVo <i>Summer Intern · Conversational AI Tools</i>	<i>May 2025 — Jul 2025</i>
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- Built PersonaBot, a configurable multi-agent orchestration toolkit with a state-managed persona system spanning 6 profiles and 10 customizable traits.
- Implemented persistent, multi-session conversation state across four languages (English, Sinhala, Tamil, Hindi), enabling bulk generation of 100+ Q&A pairs.

Blue Robins <i>Tutor · Project-Based CS Education (Grades 3–12, US students)</i>	<i>Starting [MONTH YEAR] · part-time</i>
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- Design project-based software curricula across grade bands, from lightweight app ideas to mentor-led builds in ray tracing, Naive Bayes, pathfinding, and neural networks.

- Grounded a Universal Access Engine curriculum in HCI theory — Norman’s model, affordances, universal design — structured as theory-heavy sessions feeding a 3-week build phase.

PROJECTS

Android Malware Detection with GNNs

Jan 2026 — May 2026

Cybersecurity · Graph ML

- Built an end-to-end malware classification pipeline extracting Function Call Graphs via Androguard and training GraphSAGE models to distinguish benign from malicious Android apps.
- Designed an API-aware, structure-preserving graph pruning technique that removes low-signal non-API methods via weighted edge rewiring.
- Integrated pretrained Skip-Gram node embeddings with checkpoint-based training for resumable HPC runs.
- Achieved a 94.39% F1-score and 92.41% accuracy on an imbalanced dataset of 150K benign and 50K malicious samples.

GraphSAGE, Androguard, PyTorch

Finance Tracker & Analyser

Jun 2025 — Jul 2025

AI-Driven Financial Analytics

- Designed a multi-format analysis and report-generation service backed by a 2-agent pipeline for automated insight generation, since restored to a fuller multi-agent CrewAI pipeline with live per-stage timers.
- Implemented real-time budget tracking with configurable spending categories and an AI-driven recovery-strategy module.
- Built a rules-based anomaly-detection service that flags inconsistencies in uploaded statement data.

Next.js, CrewAI, Vercel

Automated Research & Ideation Pipeline

Jun 2025

Multi-Agent Systems · Async Automation

- Designed a 5-agent orchestration workflow automating topic selection, research, and outline generation with shared agent state.
- Implemented concurrent agents over Python asyncio to generate 7 niche titles per theme, each with sourced research summaries and outline guides.
- Cut research-and-ideation turnaround by ~80% (5 days → 1 session); adopted as the team’s standard ideation tool.

Python, asyncio, Multi-Agent

Tag Atlas

Personal project

Project write-up storage & tagging tool

- Sidebar-plus-detail-panel workspace for storing and tagging project write-ups, with tag-based similarity ranking and an archive flow.
- AI-assisted tag suggestions via Gemini, password auth, and bulk project-insertion tooling.

Next.js, Supabase, Gemini

LEADERSHIP

ACM — BPHC Chapter

Nov 2024 — Jul 2026

Content & Research Team Co-Lead, Junior Office & Chair, Senior Office

- Authored 5+ technical articles for the ACM BPHC Blog and The BITSian Gazette (Aug–Nov ’24 edition).
- Coordinated logistics as Point of Contact for 4 tech events, including a coding workshop for 70+ indigent students.

ELAS, BPHC

Dec 2024 — Jan 2025

ARG Team Lead, Verba Maximus ’25

- Led a 13-member team to design and execute a 6-hour event that drew 125+ participants — the highest-footfall headliner event at VM '25.

TALKS \{\}& WORKSHOPS

Introduction to Machine Learning

Aug 2025

Organized by ACM — BPHC Chapter

- Co-hosted a 250+ attendee workshop covering supervised/unsupervised learning, regression, neural networks, and activation functions with live coding.